

Aggregate (cost=18833.14..18833.15 rows=1 width=64) (actual time=173.019..175.819 rows=1.00 loops=1)
 Buffers: shared hit=9444
 -> WindowAgg (cost=18799.18..18818.58 rows=971 width=30) (actual time=172.963..175.798 rows=99.00 loops=1)
 Window: w2 AS (ORDER BY (count(*)), m.song_title ROWS UNBOUNDED PRECEDING)
 Storage: Memory Maximum Storage: 17kB
 Buffers: shared hit=9444
 -> Sort (cost=18799.16..18801.59 rows=971 width=22) (actual time=172.930..175.735 rows=99.00 loops=1)
 Sort Key: (count(*)), m.song_title
 Sort Method: quicksort Memory: 29kB
 Buffers: shared hit=9444
 -> WindowAgg (cost=18731.58..18750.98 rows=971 width=22) (actual time=172.822..175.661 rows=99.00 loops=1)
 Window: w1 AS (ORDER BY (count(*)), m.song_title ROWS UNBOUNDED PRECEDING)
 Storage: Memory Maximum Storage: 17kB
 Buffers: shared hit=9441
 -> Sort (cost=18731.56..18733.99 rows=971 width=14) (actual time=172.798..175.603 rows=99.00 loops=1)
 Sort Key: (count(*)) DESC, m.song_title
 Sort Method: quicksort Memory: 28kB
 Buffers: shared hit=9441
 -> Finalize GroupAggregate (cost=18390.66..18683.38 rows=971 width=14) (actual time=172.649..175.561 rows=99.00 loops=1)
 Group Key: m.song_title
 Buffers: shared hit=9441
 -> Gather Merge (cost=18390.66..18662.02 rows=2330 width=14) (actual time=172.643..175.504 rows=297.00 loops=1)
 Workers Planned: 2
 Workers Launched: 2
 Buffers: shared hit=9441
 -> Sort (cost=17390.63..17393.06 rows=971 width=14) (actual time=165.784..165.793 rows=99.00 loops=3)
 Sort Key: m.song_title
 Sort Method: quicksort Memory: 28kB
 Buffers: shared hit=9441
 Worker 0: Sort Method: quicksort Memory: 28kB
 Worker 1: Sort Method: quicksort Memory: 28kB
 -> Partial HashAggregate (cost=17332.74..17342.45 rows=971 width=14) (actual time=165.705..165.724 rows=99.00 loops=3)
 Group Key: m.song_title
 Batches: 1 Memory Usage: 65kB
 Buffers: shared hit=9425
 Worker 0: Batches: 1 Memory Usage: 65kB
 Worker 1: Batches: 1 Memory Usage: 65kB

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-> Hash Join (cost=2905.03..16608.74 rows=144801
width=6) (actual time=17.593..142.115 rows=116176.00 loops=3)
    Hash Cond: (p.song_id = m.song_id)
    Buffers: shared hit=9425
-> Parallel Hash Join (cost=2433.03..15756.48
rows=144801 width=4) (actual time=12.447..112.863 rows=116176.00 loops=3)
    Hash Cond: (p.user_id = u.user_id)
    Buffers: shared hit=8684
-> Parallel Seq Scan on playlists p
(cost=0.00..12561.33 rows=290317 width=20) (actual time=0.030..39.913 rows=232246.33
loops=3)
    Filter: (added_date > '2023-01-01'::date)
    Rows Removed by Filter: 101087
    Buffers: shared hit=7353
-> Parallel Hash (cost=2066.29..2066.29
rows=29339 width=16) (actual time=12.237..12.237 rows=16656.67 loops=3)
    Buckets: 65536 Batches: 1 Memory Usage:
2912kB
    Buffers: shared hit=1331
-> Parallel Seq Scan on users u
(cost=0.00..2066.29 rows=29339 width=16) (actual time=0.013..7.248 rows=16656.67
loops=3)
    Filter: ((subscription_type)::text =
'premium'::text)
    Rows Removed by Filter: 16677
    Buffers: shared hit=1331
-> Hash (cost=347.00..347.00 rows=10000 width=10)
(actual time=5.074..5.074 rows=10000.00 loops=3)
    Buckets: 16384 Batches: 1 Memory Usage: 560kB
    Buffers: shared hit=741
-> Seq Scan on music m (cost=0.00..347.00
rows=10000 width=10) (actual time=0.070..2.555 rows=10000.00 loops=3)
    Buffers: shared hit=741

Planning:
    Buffers: shared hit=104 read=9 dirtied=2
Planning Time: 2.388 ms
Execution Time: 175.986 ms

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